



In the name of Allah, the most beneficent and the most merciful

Lecture 1

# Hydroinformatics: Data Management and Analysis

IWRM-526 & HID-526

M.S. in IWRM/HID

## Introduction to Hydroinformatics

Course Lead: Dr. Arjumand Zaidi

Instructor: Mr. Ibraheem

USPCAS-W, MUET, Jamshoro



# INTRODUCE OURSELVES

## WE

Dr. Arjumand Zaidi and Mr. Ibraheem

## YOU

- Name and city
- Undergraduate degree and university
- Job experience
- Research interest
- Your career expectation after masters

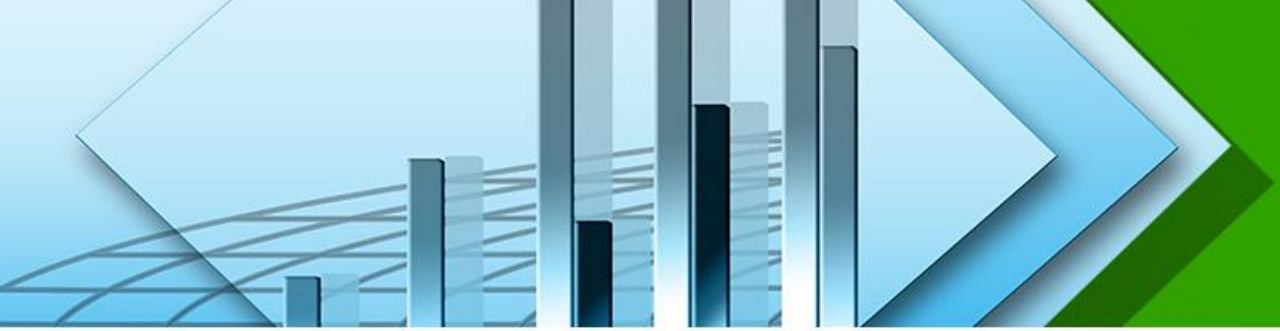
## 2+1 Credit Hours

1. Mid Exams
2. Final Exams
3. Sessional
4. Lab/Project



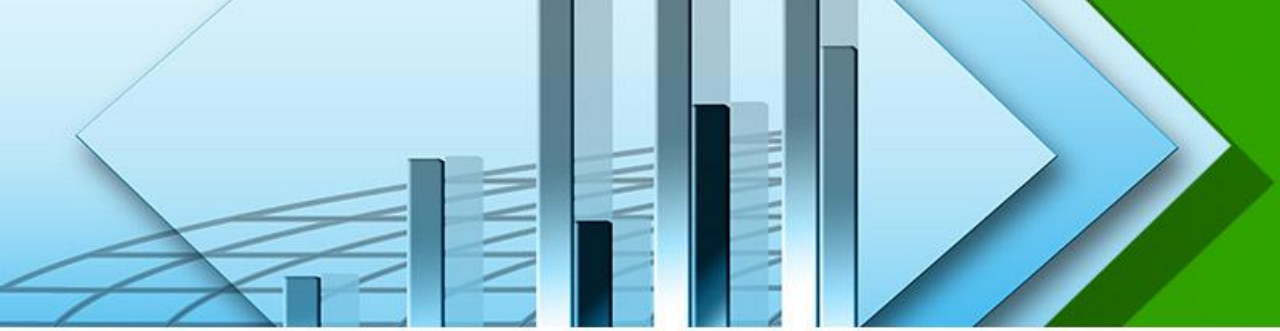
# ACCESS THE INSTRUCTOR

- For any help regarding exercises, research projects you can reach out to us:
  - Access via email
  - Open office hours
  - Office Hours: Any time by appointment



# US-PCAS W POLICY

- **Punctuality and Regularity** will be appreciated in the class.
- **Schedule of Assignments:** Assignments will be given to students throughout the semester with well-defined due dates.
- Students who **missed the submission deadline** of exercise will not have chance to submit the exercise.
- The **90% attendance requirement** is also necessary to qualify for appearing in the final examination of the semester. However, a condonation in attendance to the extent **of 15% for** the purpose of appearing in the final semester exam can be considered only in genuine cases and under unusual circumstances.
- **Disability accommodations:** University provides an equal educational opportunity to students with disabilities otherwise qualified and equal access to University programs and activities provided to other students.



# US-PCAS W POLICY

- **Policy on Academic Honesty:** All work submitted by students should be their own. Plagiarism of any type will not be tolerated and will result in an F grade. Students are highly recommended to attend a workshop on plagiarism, its types, and ways to avoid plagiarism.
- **Policy on Academic Misconduct/Dishonesty:** There is a zero-tolerance policy for any form of academic dishonesty. Students who engage in plagiarism, cheating, or forgery on any assignment or test will be subject to the University's conduct code policies.
- **Social Justice Statement:** "Mehran University of Engineering & Technology (MUET)" is committed to social justice. I concur with that commitment and expect to foster a nurturing learning environment based on open communication, mutual respect, and non-discrimination. Our University does not discriminate based on race, sex, age, disability, veteran status, religion, sexual orientation, color, or national origin. Any suggestions on how to further such a positive and open environment in this class will be appreciated and seriously considered.





# Learning objectives of course

At the end of this course, students will be able to:

1. Understand the principles and applications of Hydroinformatics.
2. Apply geospatial and data analysis tools to solve hydrological problems.
3. Evaluate hydrological data to support informed and sustainable decision-making.



# SEMESTER GOALS FROM THIS COURSE

1. Understand the environmental data lifecycle and data management principles.
2. Acquire, organize, manage, and visualize environmental datasets.
3. Apply GIS, remote sensing, and environmental databases for data analysis.
4. Perform statistical and spatial analysis of environmental data.
5. Develop basic programming and database skills for environmental applications.
6. Apply environmental modeling and data-driven approaches to support decision-making.
7. Interpret and communicate environmental information using appropriate visualization techniques.





# Software we will learn in this course

- Core hands-on software
  - Microsoft Excel
  - Python (Jupyter Notebook, Pandas, NumPy, Matplotlib, scikit-learn)
  - Google Earth Engine (GEE)
  - QGIS/ArcGIS Pro
- Demonstration/introductory software
  - HEC-HMS – Rainfall-runoff modeling concepts
  - HEC-RAS / GeoHECRAS – River hydraulics and flood inundation modeling
  - ~~– SWAT or SWAT+ – Watershed and water quality modeling (overview/demo only)~~
  - ~~– SNAP – Sentinel satellite data processing (selected lecture/demo)~~

# In Lectures 1 & 2, we will Revisit Undergrad basics!

- Data & Information
- Definitions: Hydroinformatics
- Applications of Hydroinformatics
- Hydro-informatics in Pakistan

# Why we need water related information?

- Monitoring and early warnings
- Optimal management
- Equitable allocation
- Reduced uncertainties
- Hazard and disaster management
- Etc.



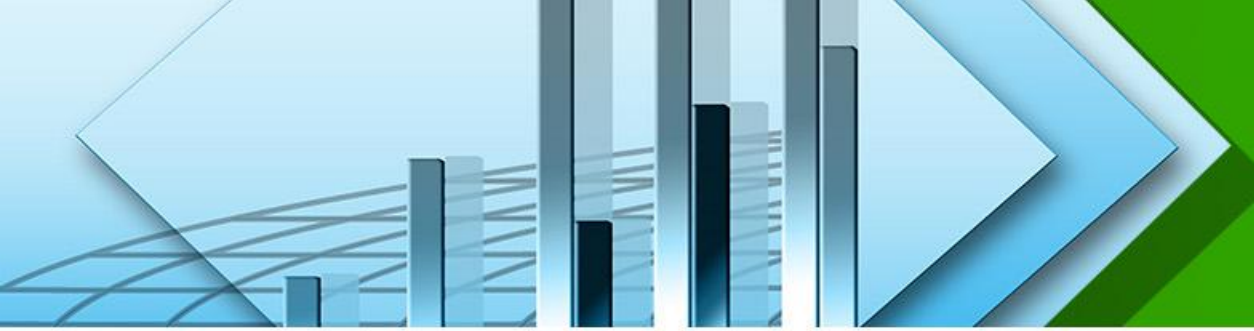
# Do you know?

- The difference between –Data and Information
- Your experience with different types of Hydro Data and how you derived information?
- Any examples from your undergraduate degree/last semester?



# Today's Mental exercise!!!

- What is data?
- Difference between Data and Information?
- What are variables?
- What is the difference between concepts and variables?



# DATA and information

- Data refers to raw facts, figures, or symbols that have not been processed or organized in any meaningful way.
- Information: data processed and analyzed to provide context, meaning, and relevance.

In short, data is the building block, while information is the transformed and structured output that enables understanding and decision-making.





## Data

Data is collection of facts, with no or minimal meaning.

It is in raw form, unprocessed and unorganized.

It is independent of information.



## Information



The processed data is called information.

It is organized and put into some context.

Information is dependent on data.

# EVOLUTION OF DATA

## Data

Collected facts in unorganized form.

## Information

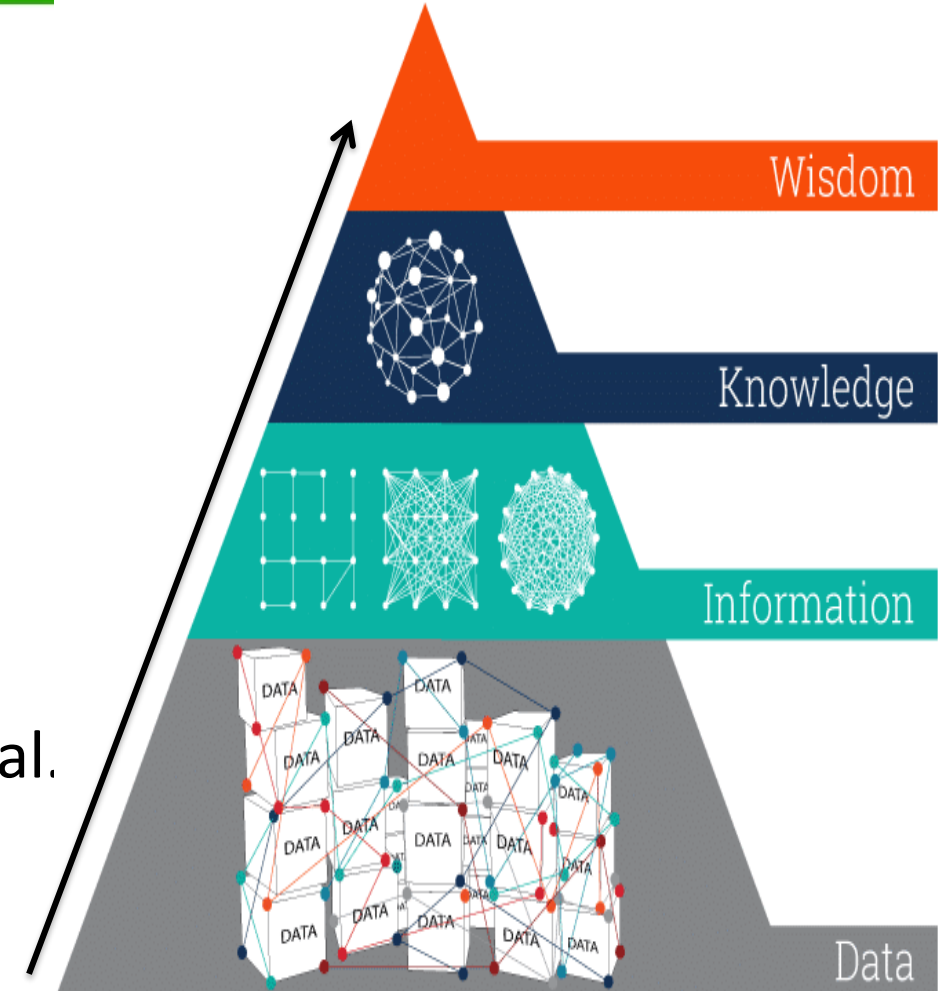
Visualized data for specific purpose.

## Knowledge

Connected Information, to apply and achieve a goal.

## Wisdom

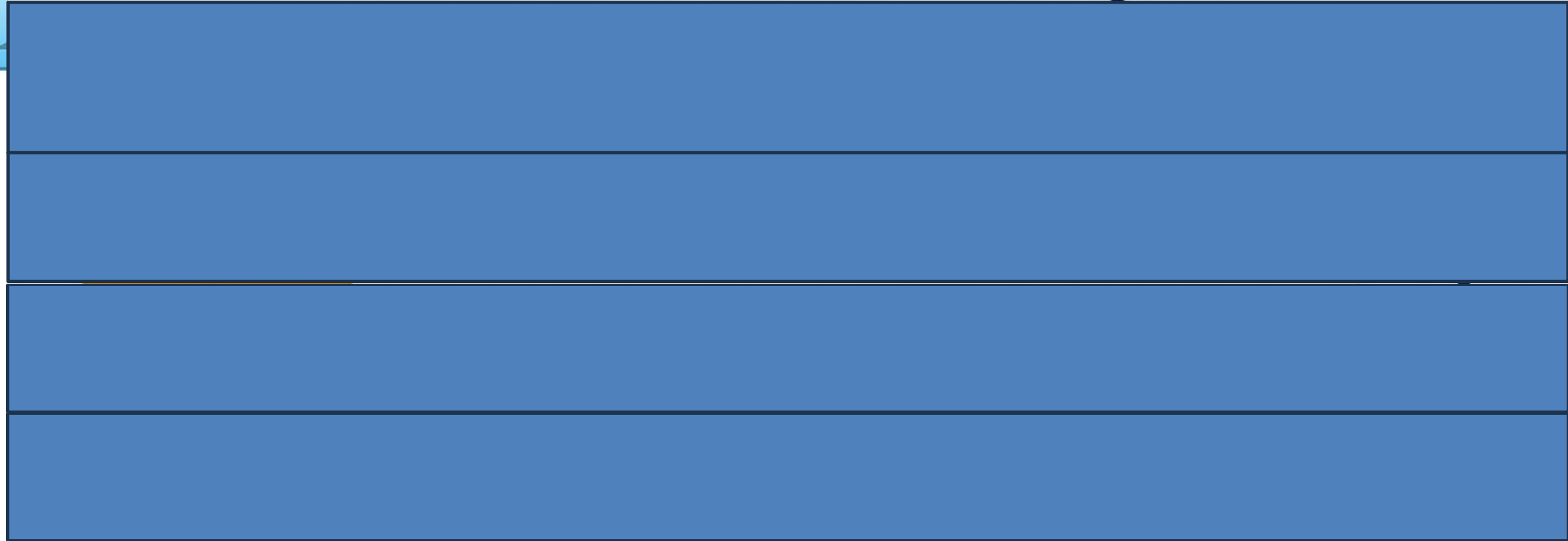
Knowledge applied in action.



DIKW Pyramid

# DIKW PYRAMID

●———— From Data to Wise Decisions in Flood Management ————●



## DATA

Raw facts and observations



## INFORMATION

Organized and processed data



## KNOWLEDGE

Understanding from experience and patterns

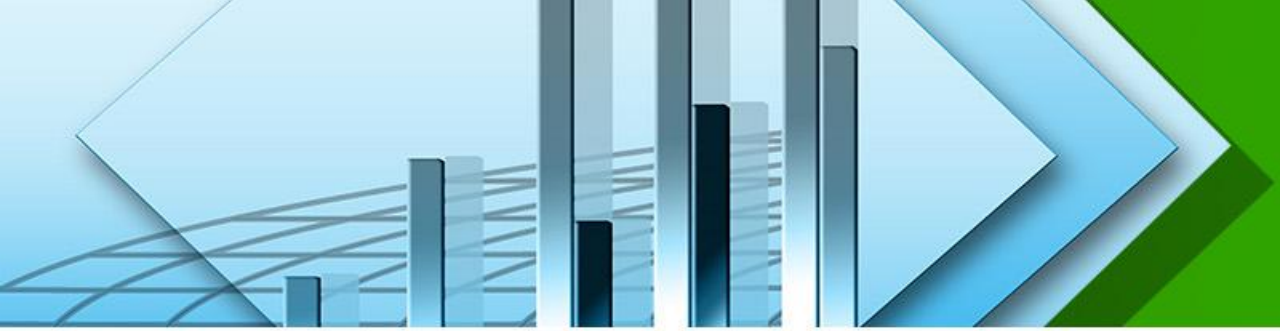


## WISDOM

Good judgment and effective action



**THE GOAL:** Turn data into wise decisions to save lives, reduce damage, and build resilience.

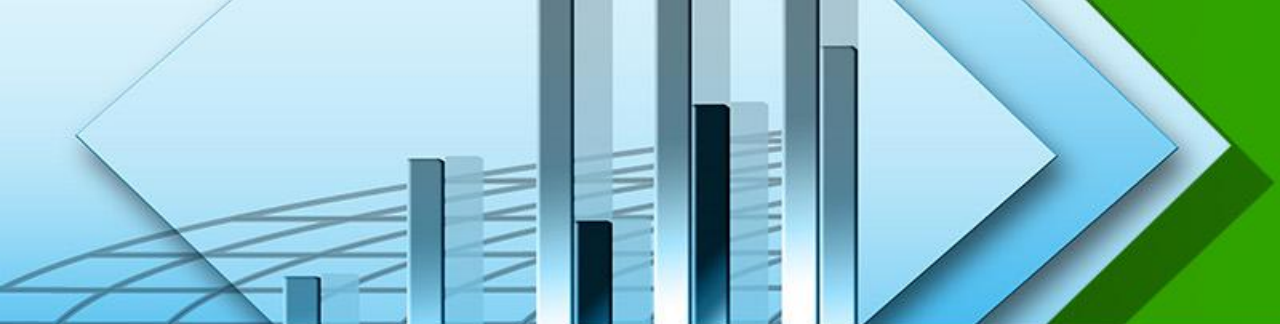


# What are concepts?

- Concepts are formed through individuals' mental images and perceptions, and their meanings vary person to person.
- Difficult to measure as they are

## Examples:

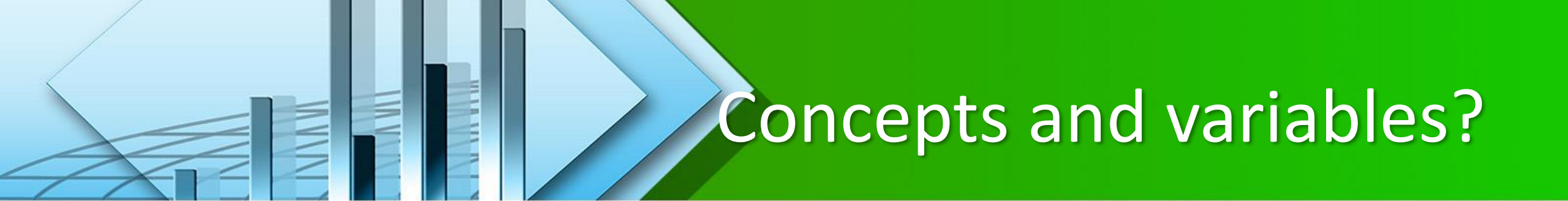
- Happiness
- Satisfaction
- Performance



# What are variables?

- A number is a measurable concept that represents quantities.
- A number is a symbol that is associated with values.
- Alternatively, a variable is an entity that exhibits variation or change (can take on different values).

Main difference between a concept and a variable is measurability



# Concepts and variables?

- Concepts should be converted into variables for measurements.
- Any concept that is capable of measurement is a variable.



# Operationalize concepts

To minimize the level of variation in people's interpretation, it is essential to operationalize concepts by transforming them into indicators and variables.





# Converting concepts into variables

- Identify indicators.
- Convert indicators into variables.
- Choices of indicators may vary with the researcher.

There should be a logical link between indicators and their concepts.

# concepts into variables



- Example 1: **Rich**
  - Find indicators for wealth
- Example 2: **High academic achievement**
  - Indicators???
- Example 3: **Powerful**
  - Indicators???



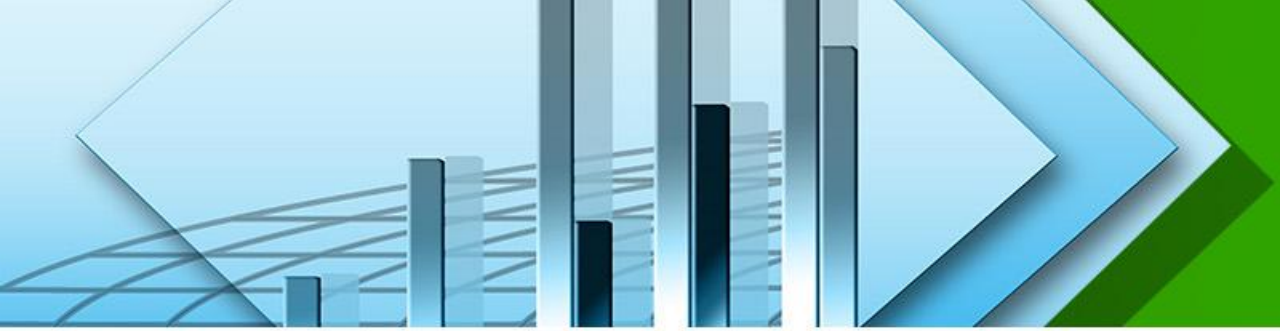
# Examples of concepts & Variables

## Concepts

- Effectiveness
- Satisfaction
- Water Quality
- Excellent academic performance
  - Rich

## Variables

- enrollments in a program
- Employees complaints
  - Bacteria count
  - GPA
  - Income



# Types of variables

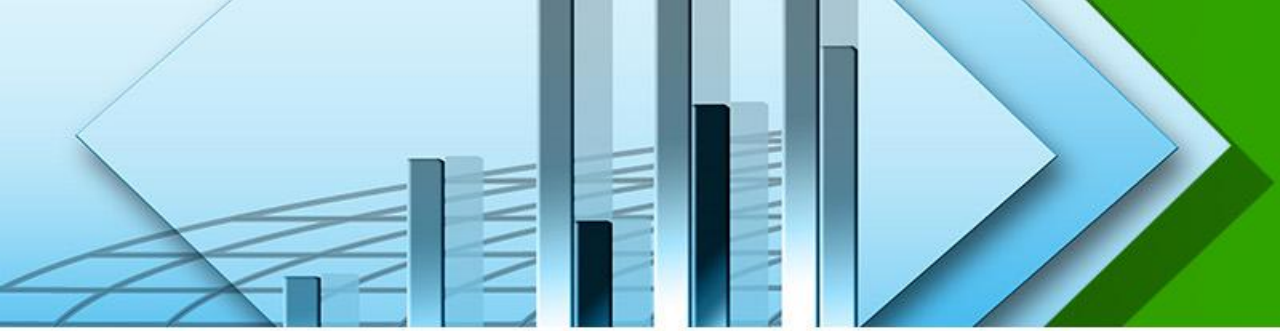
From the viewpoint of

- causal relationship
- unit of measurements



## From viewpoint of causal relationship

- Change variables or the independent variable
  - Responsible for bringing about change in a phenomenon, situation, and circumstance.
- Outcome or Dependent variable
  - Effects, impacts. Or the consequences of a change in a variable.
- Extraneous (influencing)
  - Affect change in the dependent variable.
- Intervening (linking)
  - Establish a relationship/link between independent and dependent variables.



Independent variable  
on which you have a control

Dependent variable

variation in the dependent variable depends on the variation in the independent variable

Precipitation



Runoff

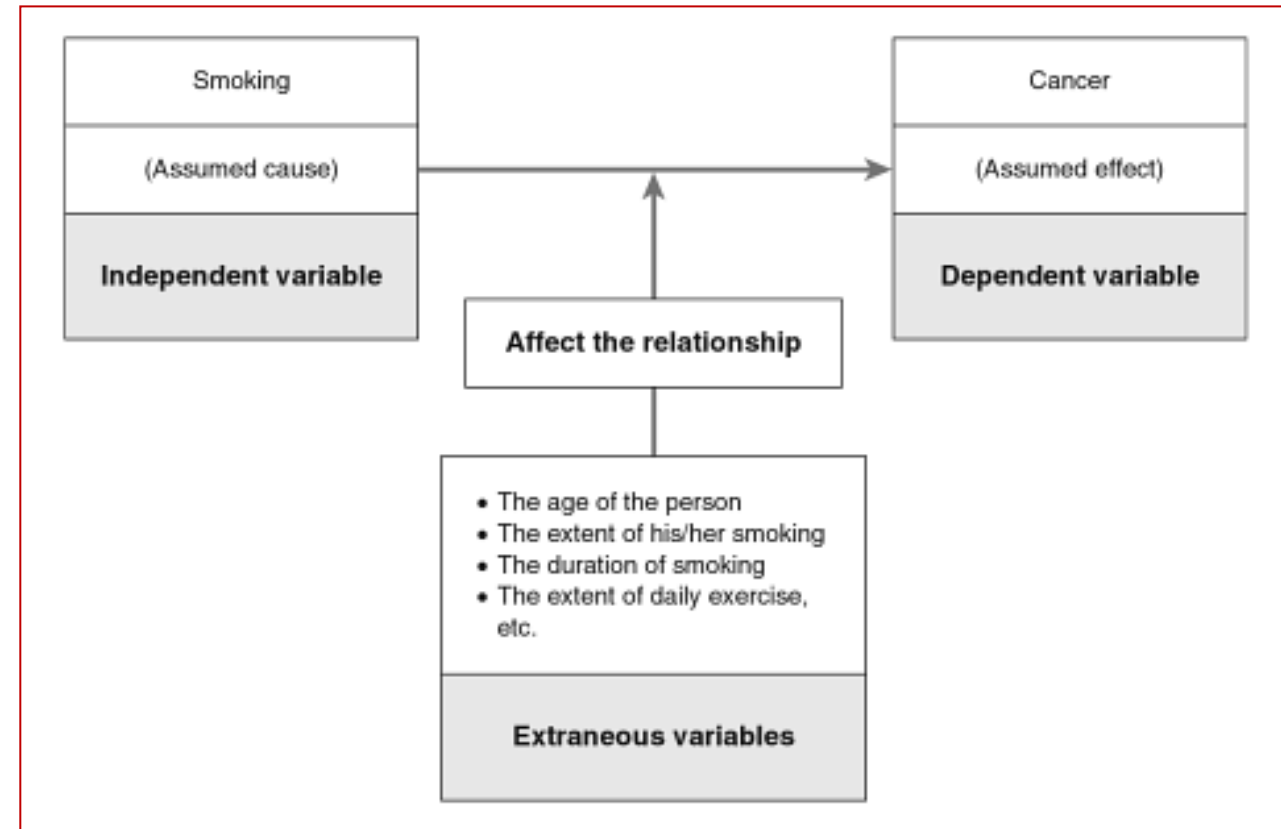
Sunny days in London



Revenue Generated

# Extraneous

- May increase or decrease the magnitude or strength of a relationship between DV and IV variables.
- Can not be controlled.

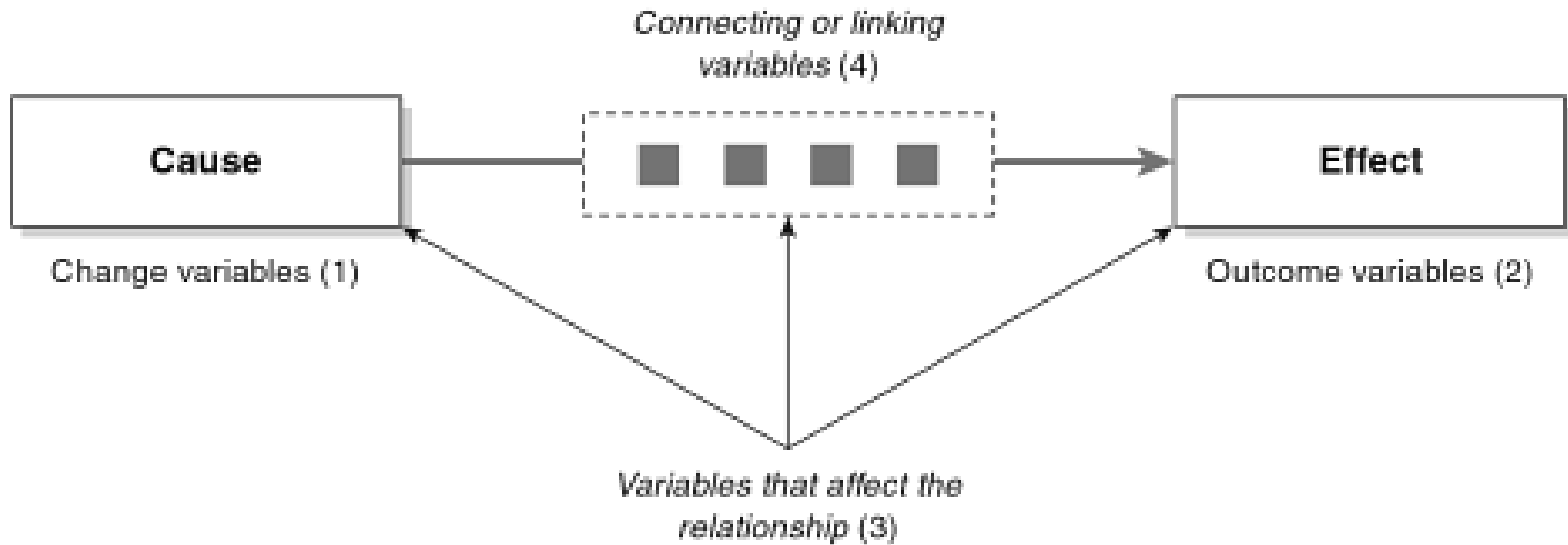


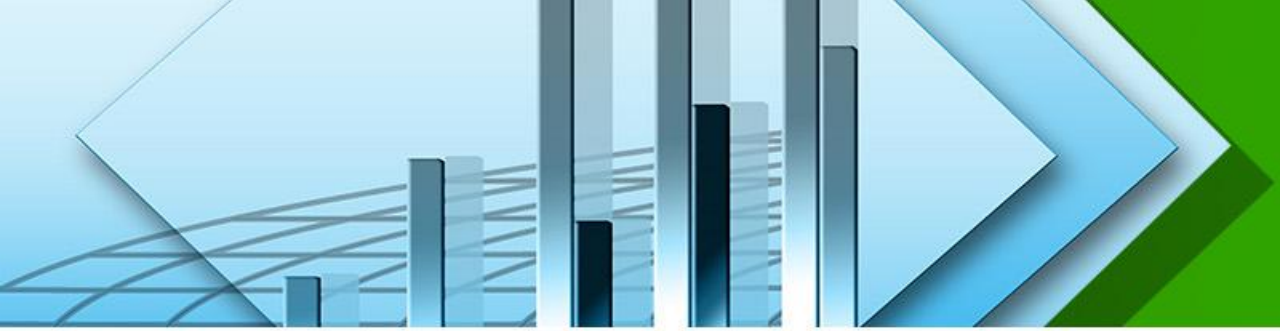




Extraneous variables are dangerous if not  
controlled !!

# Intervening





# Intervening

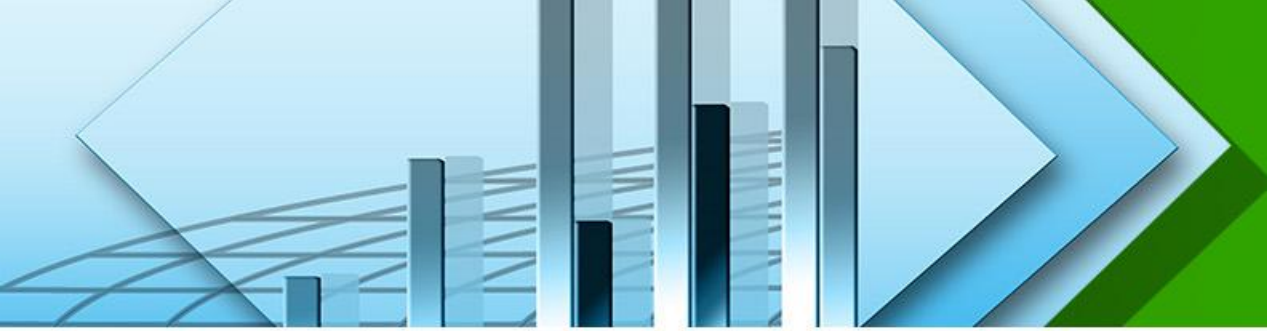
Completes the relationship between DV and IV variables

Sunny days in London

Tourism activities

Revenue Generated

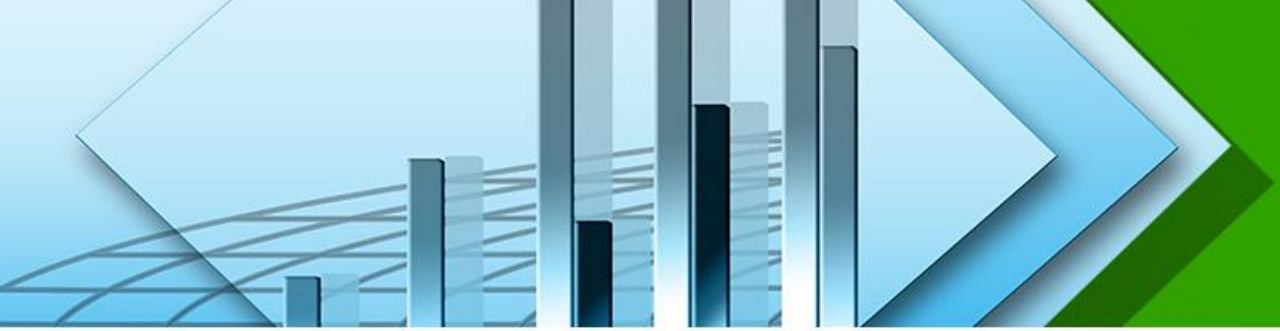
*Thus, correlation between DV and IV does NOT imply causation.*



# From viewpoint of unit of measurement

- Categorical/Continuous
- Qualitative/Quantitative

Very little difference between categorical and qualitative and between continuous and quantitative.



# Measurement Scales

## Types:

Categorical and Qualitative

- Nominal
- Ordinal

Continuous and Quantitative

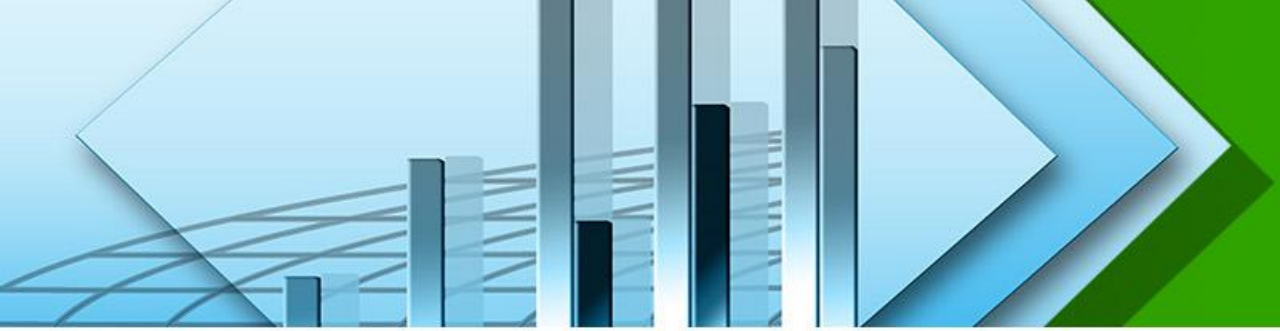
- Interval
- Ratio

- Any concept that can be measured on these scales is called a variable.
- Degree of precision in measuring a variable varies from one measurement scale to another.



# Nominal

- Each group has a characteristics which is common to all classified within that group.
- Sequence of listing of the subgroups is irrelevant since no relationship between subgroups.
- **Examples:**
  - Gender: Male/female
  - Religion: Islam/Christianity/Hindu
  - Tree, house, car, name etc.



# Ordinal

- Has characteristics of nominal scales + related subgroups.
- Arranged in ascending or descending orders.
- Ranked according to magnitude but actual magnitude is not quantifiable.
- Distance between categories is not equal, since magnitude is not quantifiable.

Example:

— Income

- Above average
- Average
- Below average
- \$ 0 – 1,999
- \$2,000-2,900
- +\$3,000

○ Age groups

- Infant
- Young
- Old



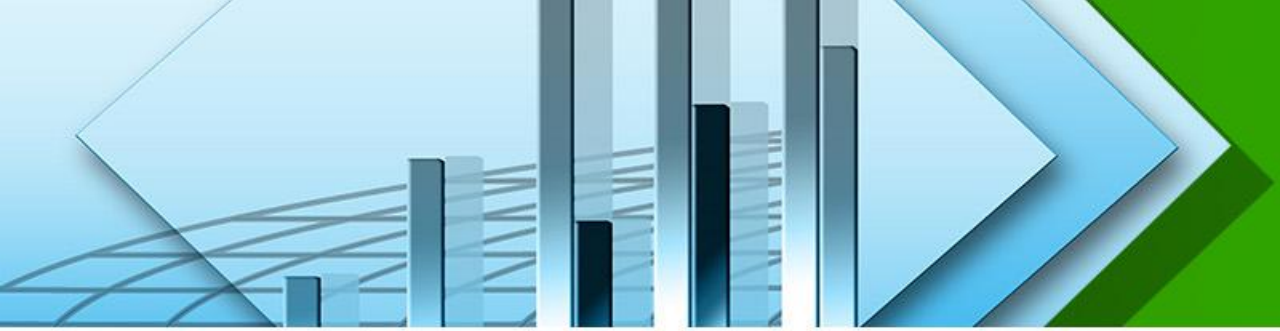


# Interval

- As an ordinal plus an arbitrary starting and ending point.

## Example:

- **Celsius Scale for measuring temperature.**
  - 0°C is the arbitrary start point with 100 °C, the arbitrary end point.
  - The gap is divided into 100 equally spaced intervals.
  - Not an absolute measurement –can't say 60°C is twice as hot as 30°C.
  - Cannot perform math calculation on each reading but can do so on differences in readings: 10°C hotter today vs. yesterday.



# Ratio

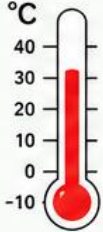
- All properties of an interval scale plus a fixed starting point, e.g., a zero point

Examples:

- Height in cm
- Flow volume
- Provides an absolute measurement – 60 cm is twice as long as 30 cm.
- Difference between intervals is the same and is measured from a zero starting point, therefore, can be used for mathematical operations.

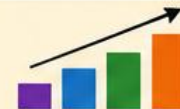
# TYPES OF VARIABLES (LEVELS OF MEASUREMENT)

Different types of variables store different kinds of information.

| 1 NOMINAL<br>(CATEGORICAL)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2 CATEGORICAL<br>(ORDINAL)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 INTERVAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 RATIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Values are <b>names or labels</b>.<br/>No inherent order.</p>                                                                                                                                                                                                                                                                                                                                                          |  <p>Categories with a <b>meaningful order</b> or ranking.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         |  <p>Numeric values with equal intervals.<br/><b>No true zero</b> (zero is arbitrary).</p>                                                                                                                                                                                                                                                                                                                                                  |  <p>Numeric values with equal intervals and a true zero.<br/><b>Zero means absence of the quantity.</b></p>                                                                                                                                                                                                                                                                                                                                               |
| EXAMPLES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | EXAMPLES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EXAMPLES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EXAMPLES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  <b>Gender:</b><br>Male, Female<br> <b>Soil type:</b><br>Clay, Loam, Sandy<br> <b>Crop type:</b><br>Wheat, Rice, Maize<br> <b>District:</b><br>Hyderabad, Dadu, Sukkur |  <b>Yield class:</b><br>Low, Medium, High<br> <b>Drought severity:</b><br>Mild, Moderate, Severe<br> <b>Farm management quality:</b> Poor, Average, Good, Excellent<br> <b>Education level:</b><br>Primary, Secondary, Higher Secondary, Graduate |  <b>Temperature in °C:</b><br>10°C, 20°C, 30°C<br> <b>Calendar year:</b><br>2000, 2010, 2020<br> <b>Time of day:</b><br>8:00 AM, 9:00 AM<br> <b>pH of soil:</b><br>5.5, 6.5, 7.5 |  <b>Rainfall (mm):</b><br>0, 50, 100, 150<br> <b>Wheat yield (kg/ha):</b><br>0, 2000, 4000<br> <b>Farm size (ha):</b><br>0, 1, 5, 10<br> <b>Irrigation water applied (mm):</b><br>0, 25, 50, 75 |
| WHAT WE CAN DO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | WHAT WE CAN DO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | WHAT WE CAN DO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | WHAT WE CAN DO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Counts, frequencies, mode<br>(most common category)                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Counts, order, median,<br>percentiles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Addition, subtraction,<br>mean, standard deviation<br>(but ratios are not meaningful)                                                                                                                                                                                                                                                                                                                                                                                                                                         | All arithmetic operations<br>including ratios<br>(e.g., twice, half)                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

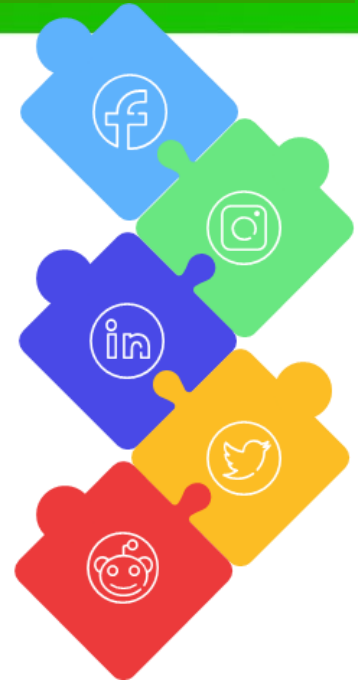


**KEY POINT:** As we move from **Nominal** → **Ordinal** → **Interval** → **Ratio**, we gain more information and can perform more complex operations.



# Why it is important to know how a variable is measured?

- Defines the type of analysis.
- Type of applicable statistical procedures.
- The way the data can be interpreted.
- Determines how the outcomes can be communicated.



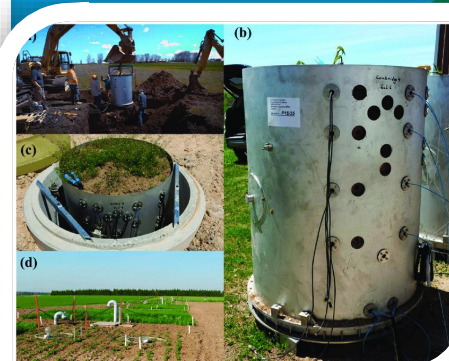
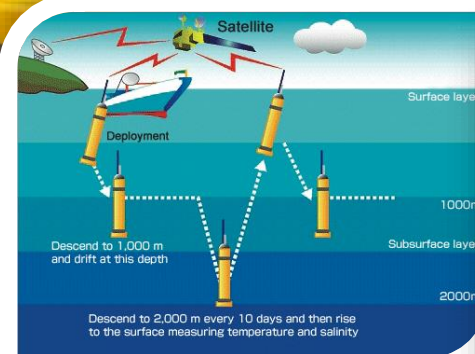
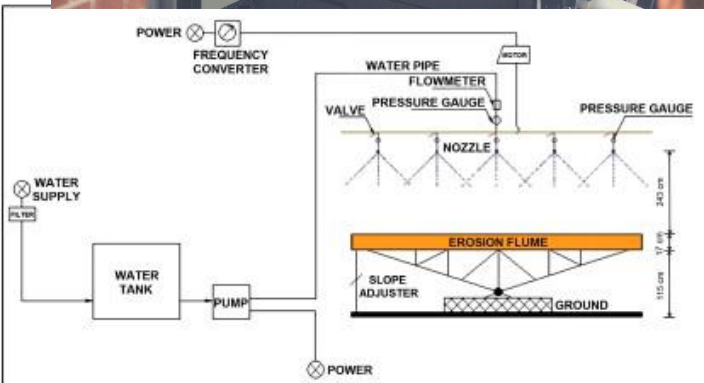
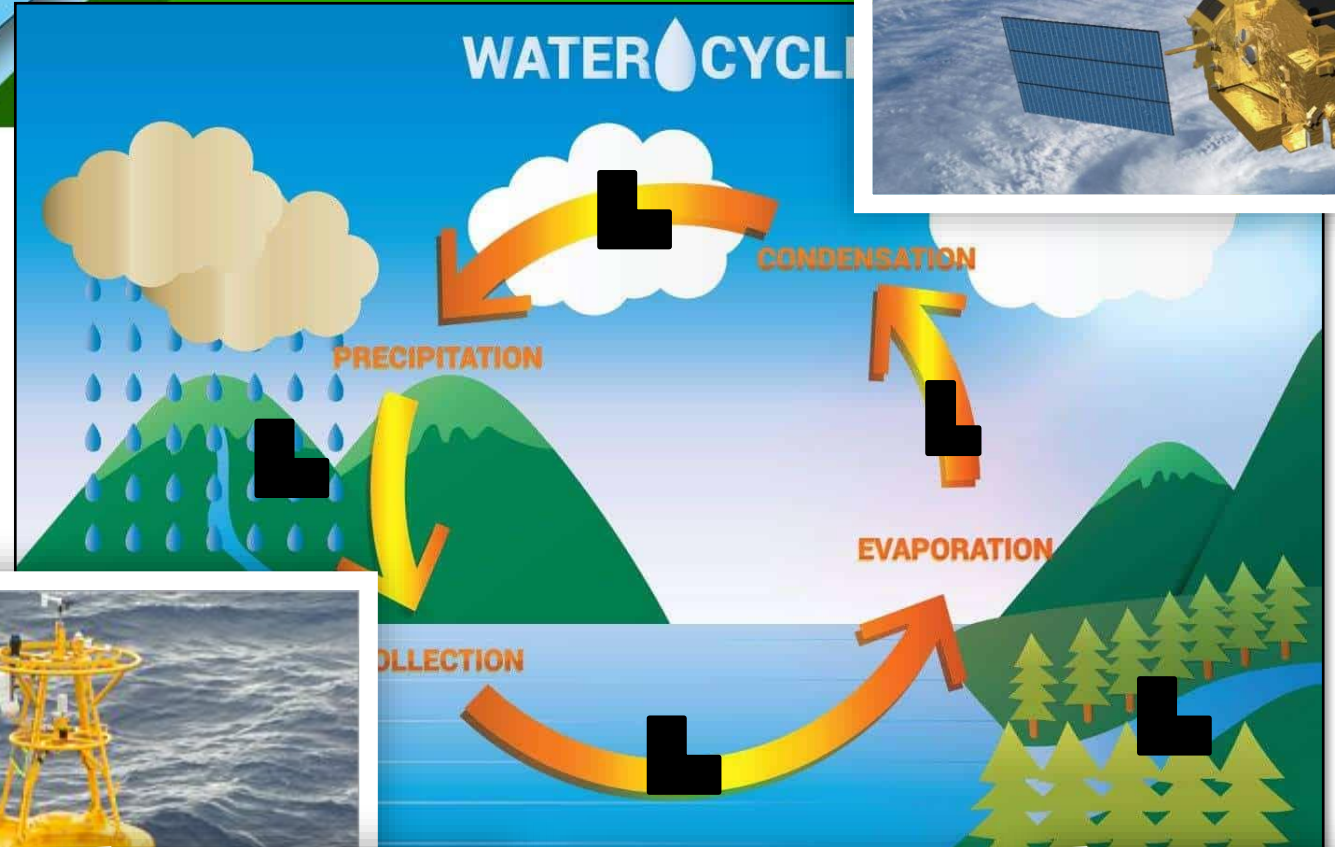
Style of your report entirely depends on the way variables are measured – way a question has been asked and its response has been recorded.



# DATA FOR HYDROINFORMATICS



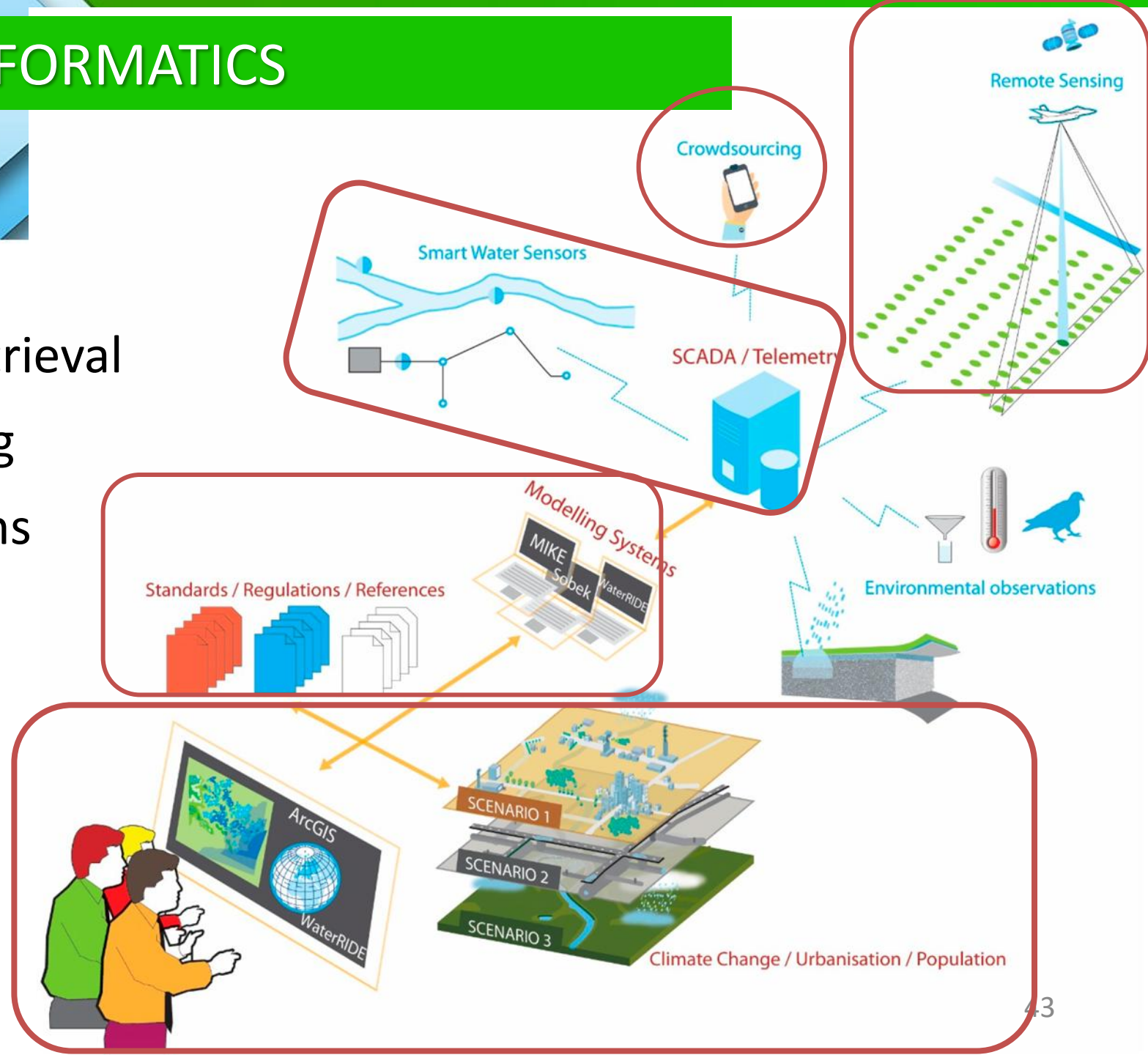
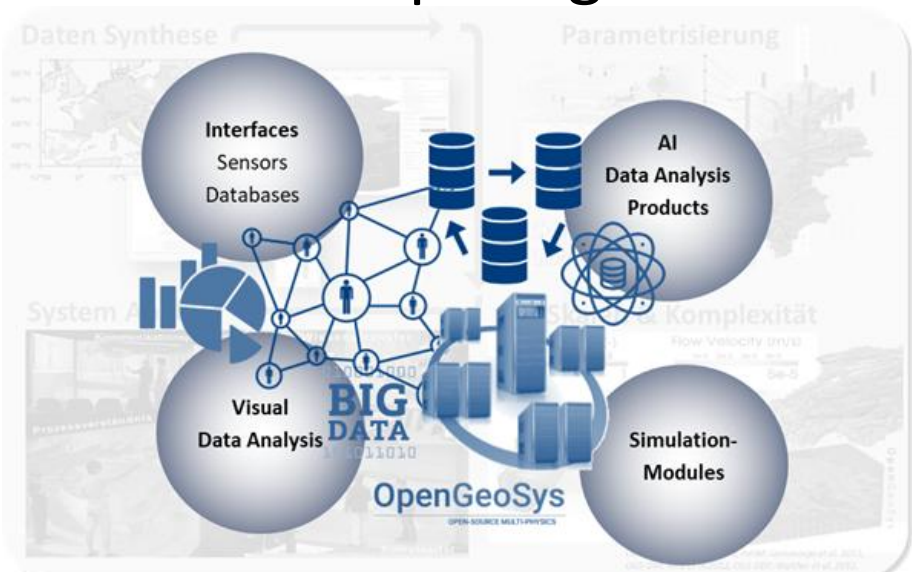
# WATER CYCLE





# TRENDS IN HYDROINFORMATICS

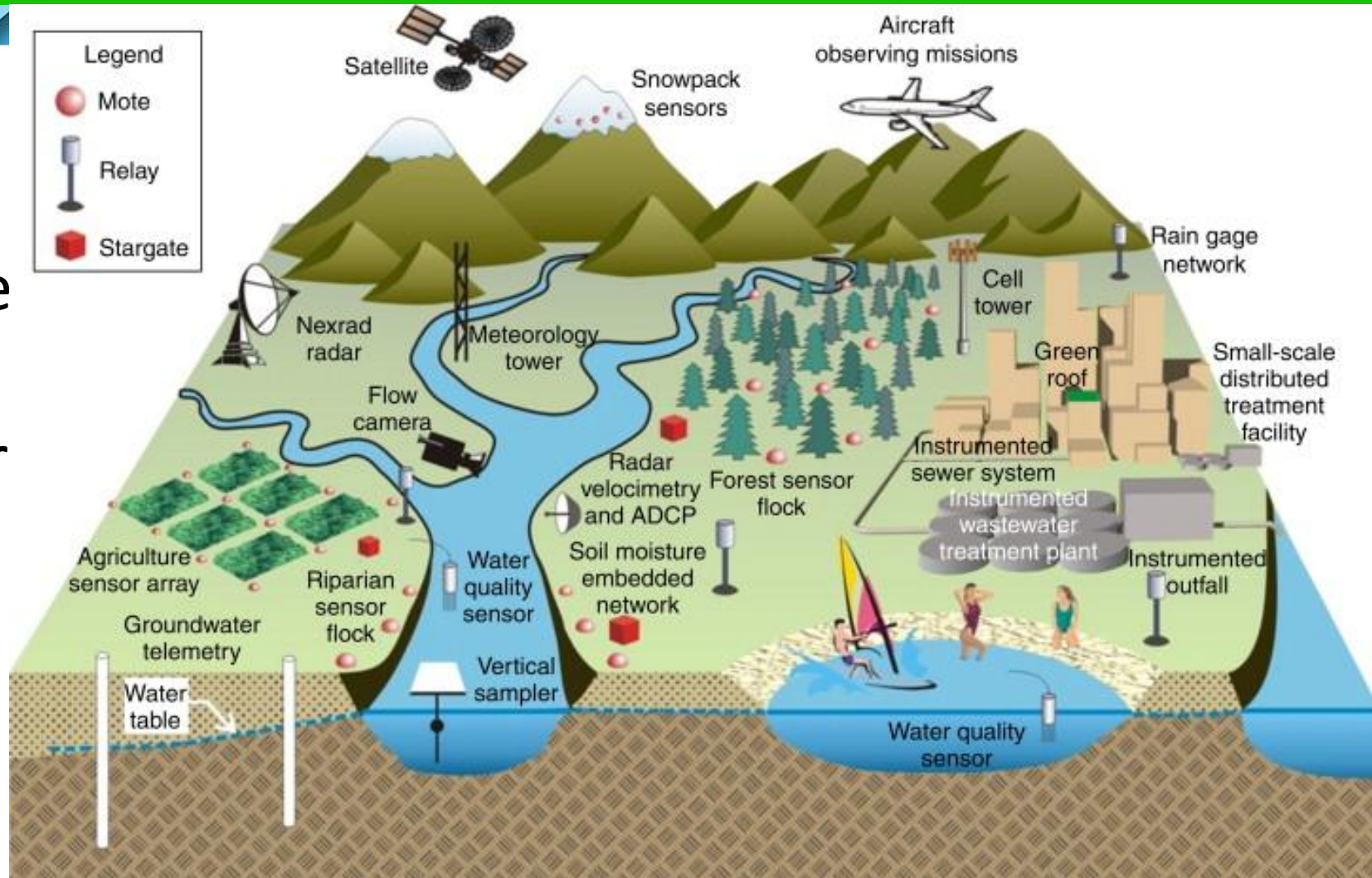
- Crowd sourcing
- Automated Sensor Data Retrieval
- Space based remote sensing
- Artificial Intelligence Systems
- Cloud computing



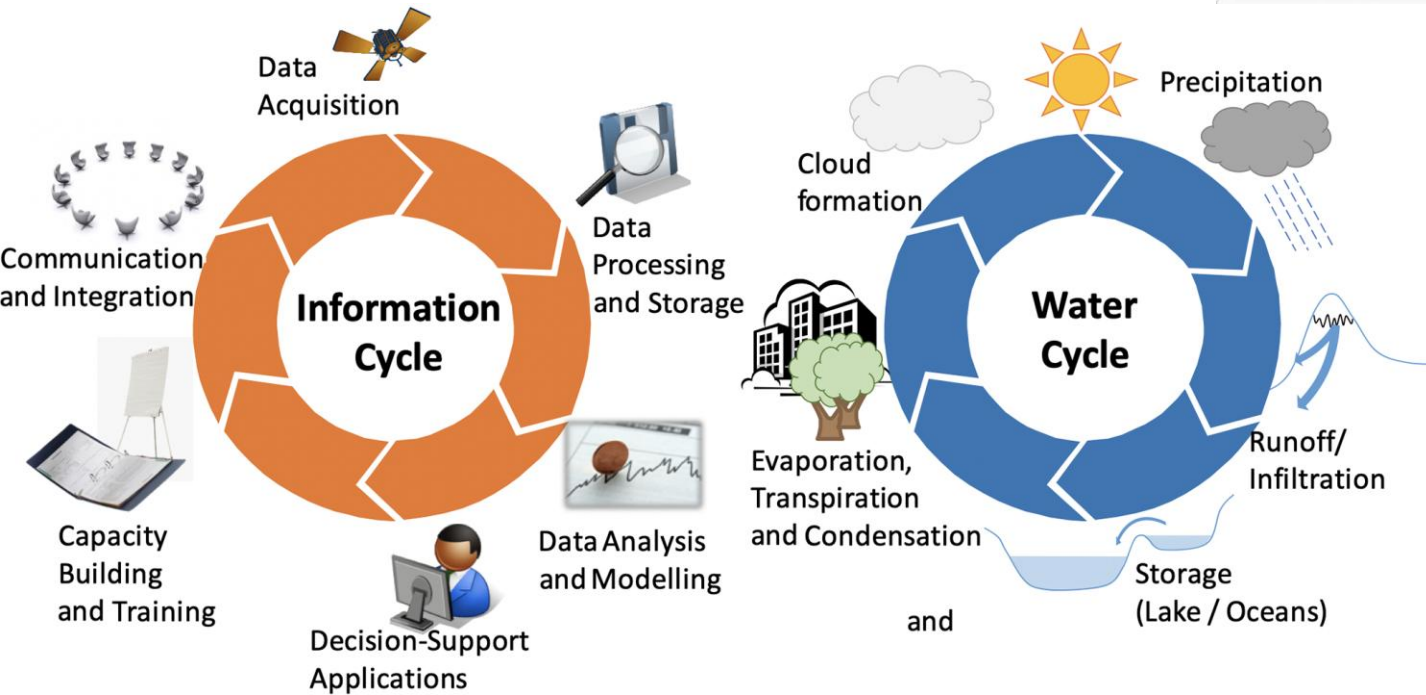
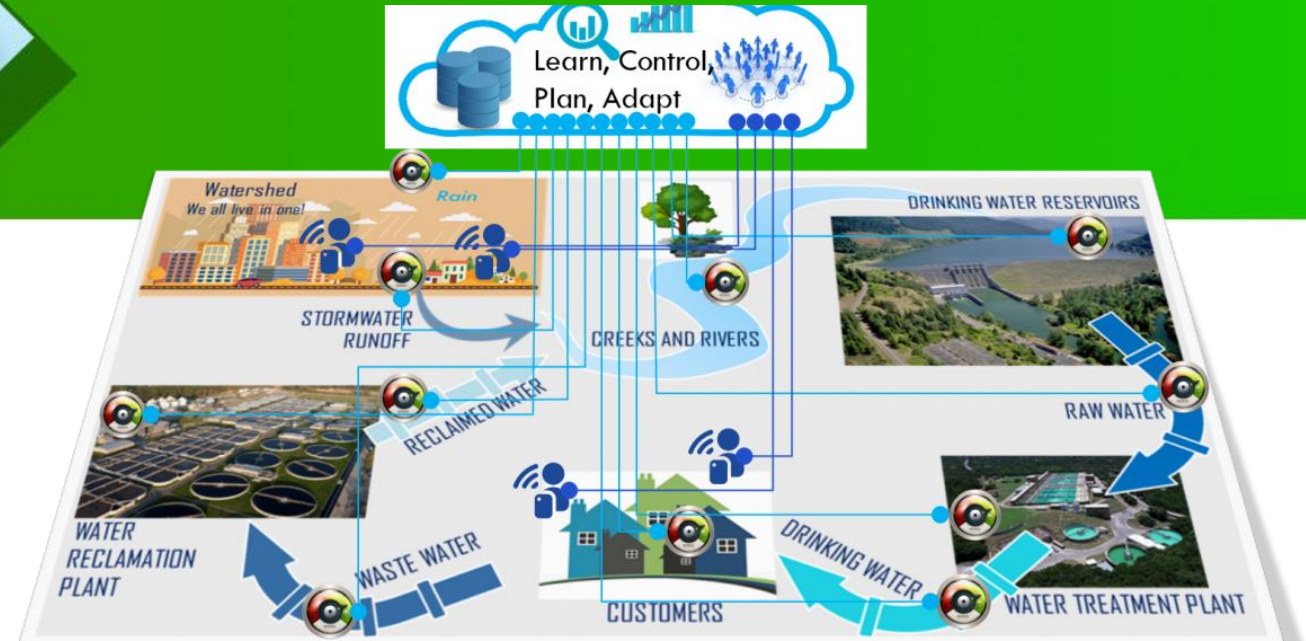


# Why we need Hydroinformatics as water professionals?

1- To build the adaptive capacity of the regions that are prone to water related disasters.



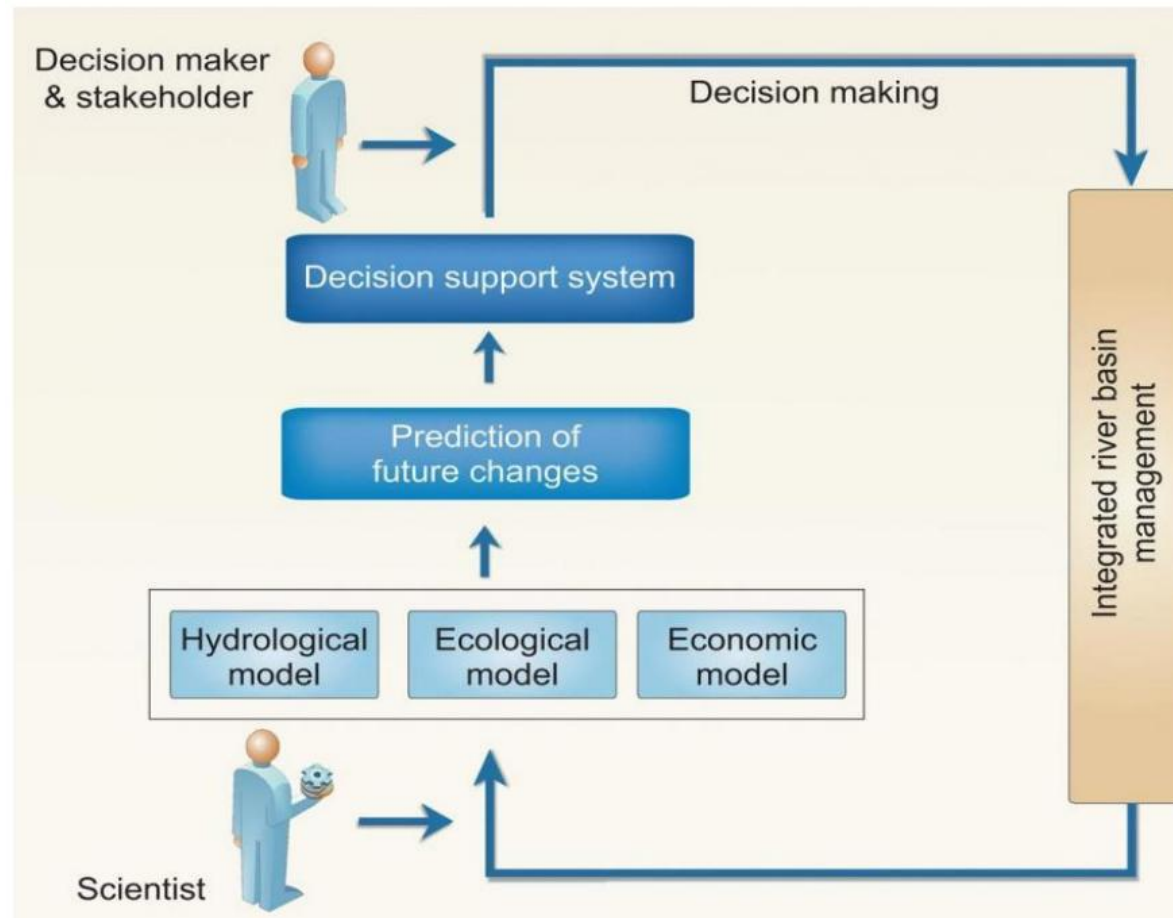
2. To track the progress of some water related intervention in natural or urban environment.



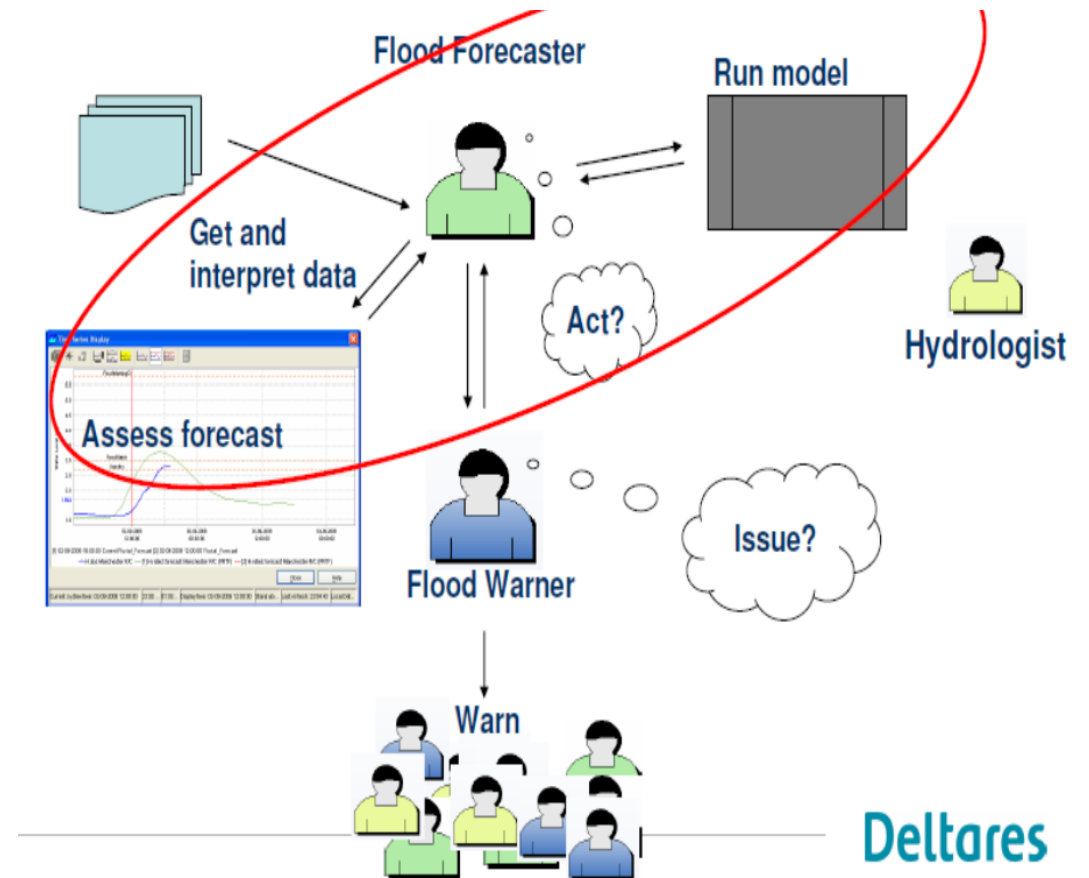
3. To practice multi-stakeholder decision making and evidence-based implementation.



4. To bring sustainable development in basin level management of rivers  
(Hydroinformatics approach to Integrated River Basin Management)



5. To solve water related conflicts among stakeholders  
E.g., Hydroinformatics in Flood Forecasting and Flood Management

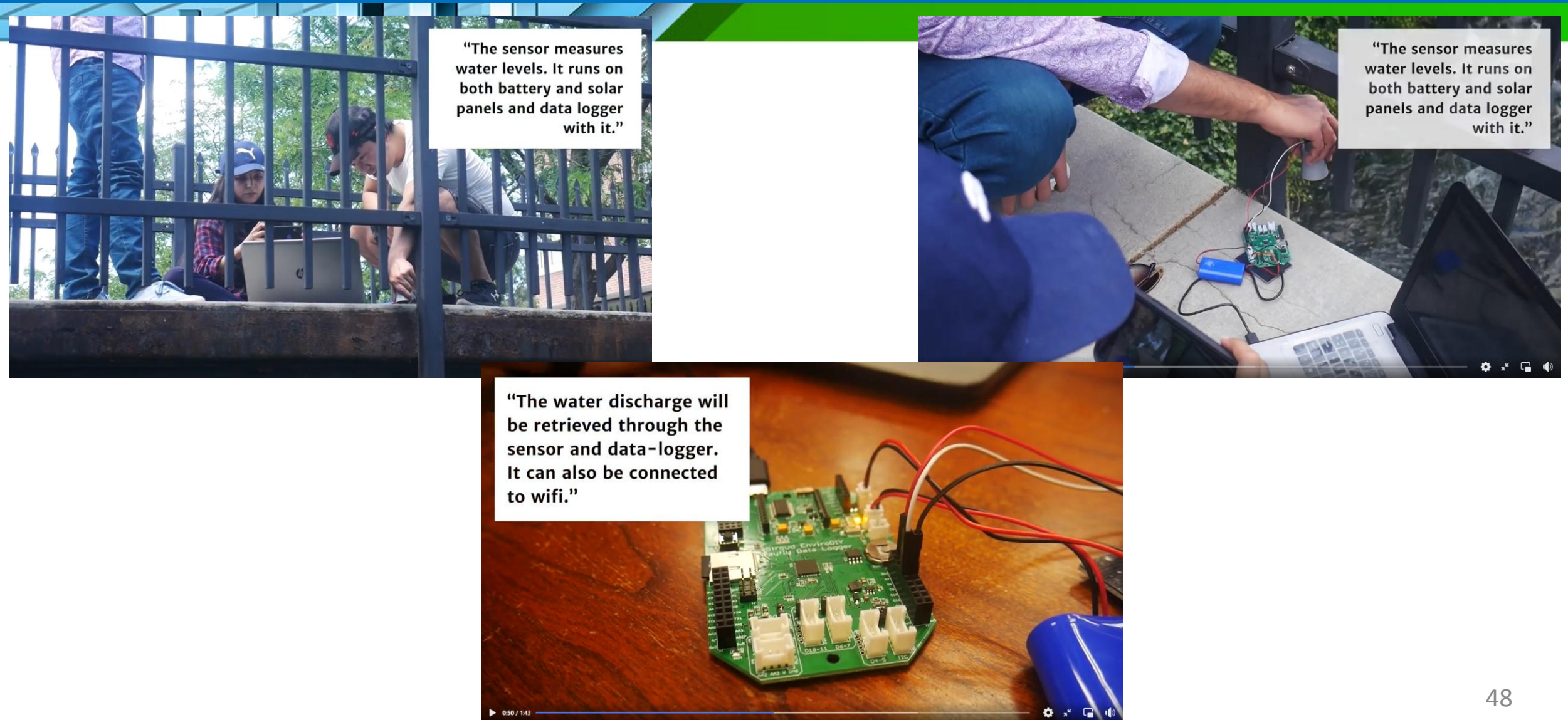




# HYDROINFORMATICS IN ACTION

## STUDENTS ENTREPRENEURS

# Development of Smart Water Sensor at University of Utah by USPCAS-W Students





# Soil Moisture Sensor & Database Development by USPCAS-W Students







# UNIVERSITY STARTUPS

# Sensor Installation for Altimetric Application at Guddu/Sukkur Barrages by USPCAS-W



# ERFMS Portal by University of Agriculture Faisalabad

[Home](#) [GPM-Based Rainfall Alerts](#) [Rivers](#) [About Us](#) [Contact Us](#)



## Extreme Rainfall and Flood Monitoring System



 Latitude

Longitude





12.13° Clear

Feels: 18.13° Low: 9.13° High: 12.09°

Partly cloudy throughout the day.

HUMIDITY

31 %

WIND

2.7 m/s

CLOUD

0%

UV INDEX

2

PRECIP

0 mm/h

VISIBILITY

16.093 km

PRESSURE

1023.6 hPa

Weekly Weather Forecast

52



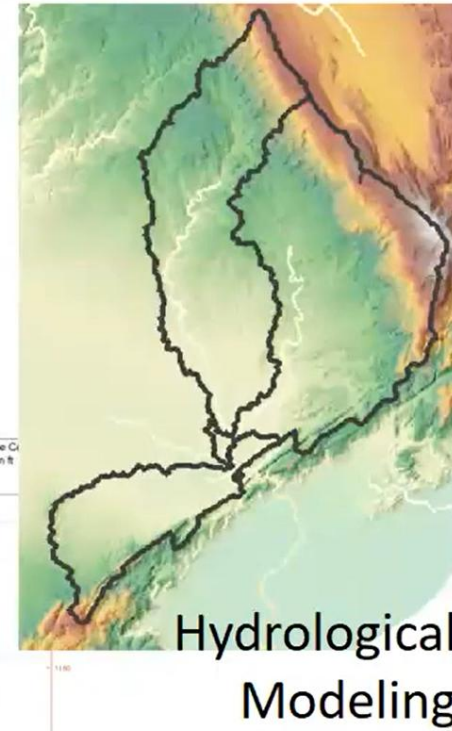
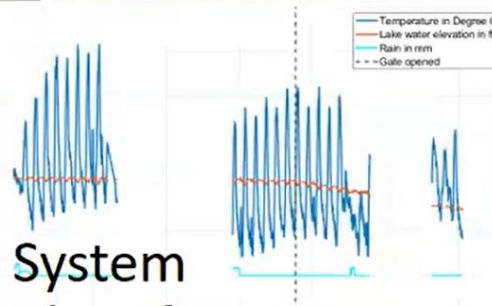
## Flood Prediction System to Prevent Social Conflict and Protect Communities



Telemetry

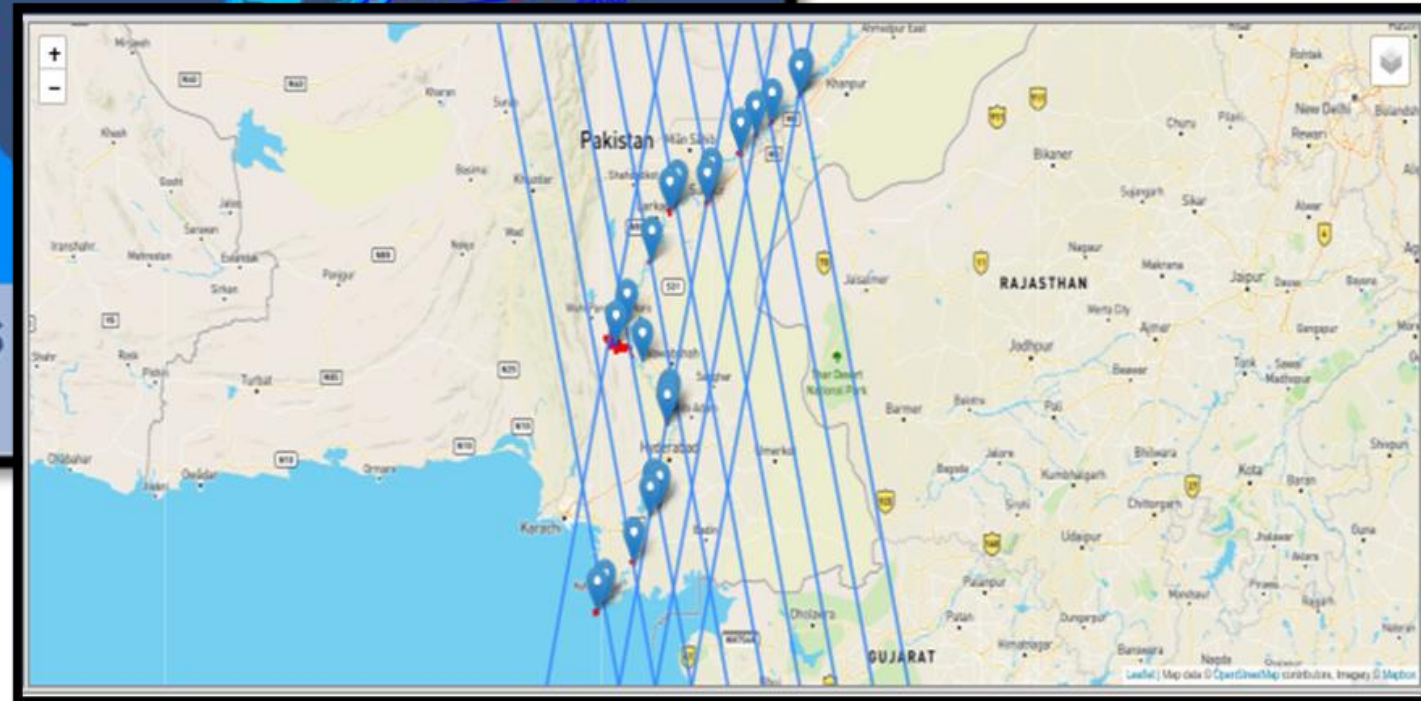
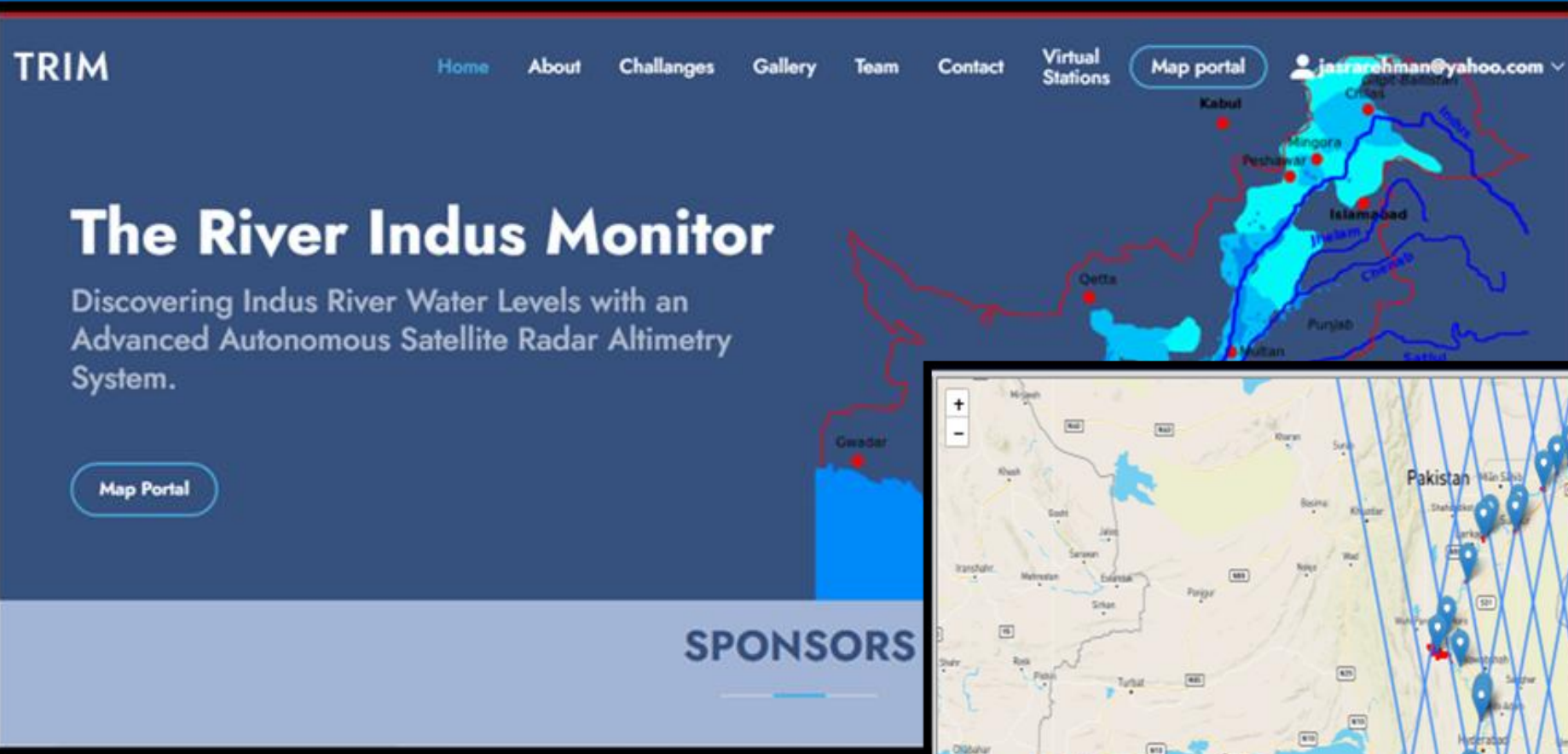


System



# TRIM – The River Indus Monitor

## USPCAS-W



<https://trim-uspcasw.vercel.app/>

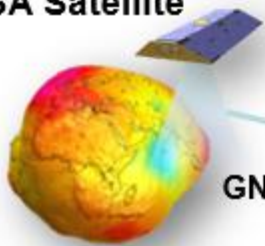




# GOVERNMENT INITIATIVES

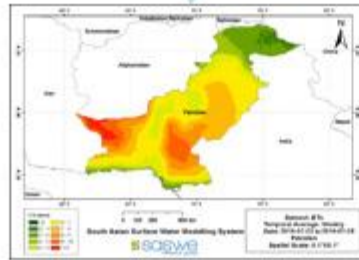
# Irrigation Advisory System by PCRWR

NASA Satellite



GNWPM

Server at PCRWR



$ET_0$   
(Evapotranspiration)

Weather  
trends/forecast  
from NASA and  
PMD

Delivery of Text  
message to farmer  
through Jazz portal



Computation and text message  
design

کسان دوست آپ کے علاقے  
ہیواپور میں پچھلے ہفتے (2 تا  
8 جولائی) میں فصلوں نے (ک  
پاس 2.5 تا 0.9 انچ) پانی  
استعمال کیا، مزید برآں اگلے  
ہفتے میں بارش متوقع نہیں  
ہے۔ لہذا اس کے مطابق آبپاشی  
کریں۔ مزید معلومات کے لیے  
03011173756 پر رابطہ  
کریں۔ شکریہ!



Crop-wise water consumption  
from 04 lysimeter stations of  
PCRWR



Dear farmer friend, we would like to  
inform you that the irrigation need  
for your banana crop was 2 inches  
during the past week.



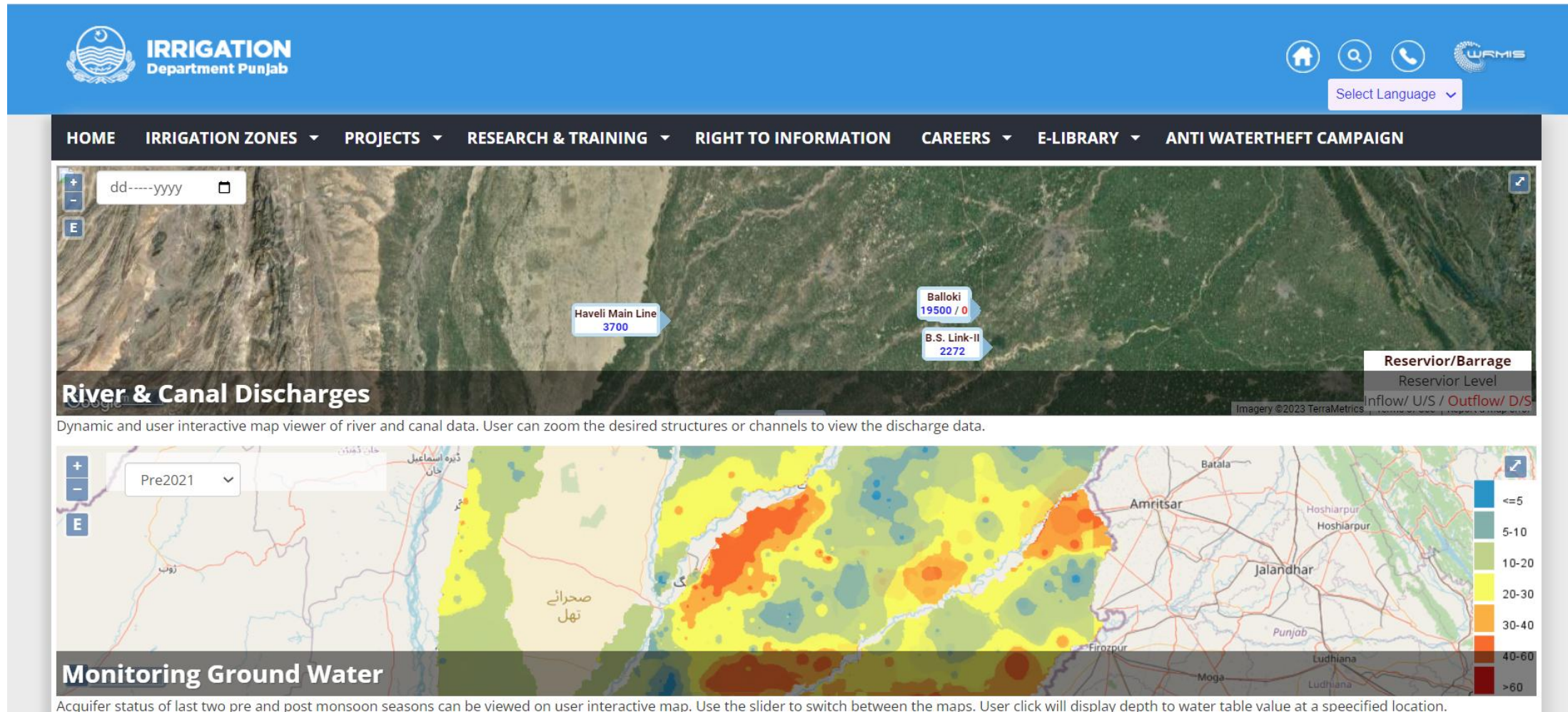
MAY 2018



SEP 2018

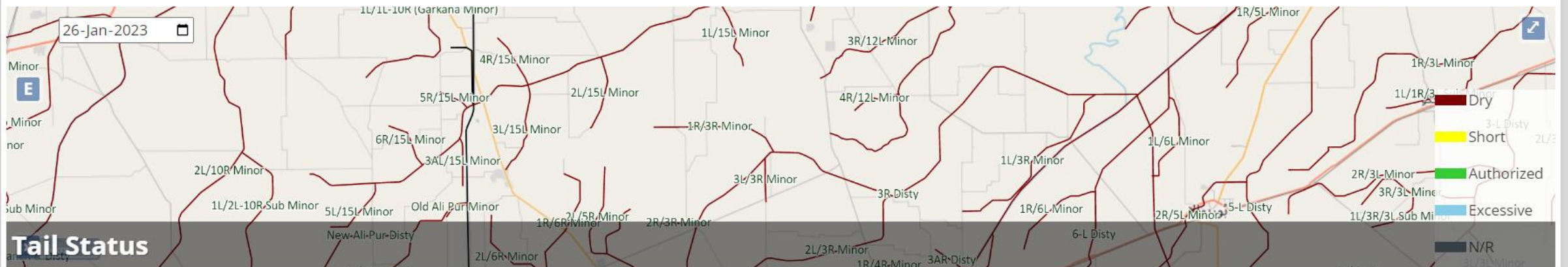


# Surface and Groundwater Discharge Portal by Punjab Irrigation Department





# Rotational Program and Tail Watercourse by Punjab Irrigation Department





# Pakistan Metrological Department Daily Database



**Flood Forecasting Division (FFD), Lahore**  
Pakistan Meteorological Department

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[PMD Website](#)

[Internship Program](#)

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## INDUS RIVER (TARBELA DAM)

Tarbela Reservoir Daily Mean Flows (Cusecs) on 27-January-2023



Daily Tarbela Reservoir Level (Feet) on 27-January-2023

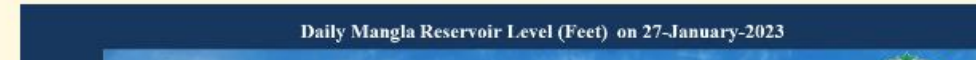


## JHELM RIVER (MANGLA DAM)

Mangla Reservoir Daily Mean Flows (Cusecs) on 27-January-2023



Daily Mangla Reservoir Level (Feet) on 27-January-2023



## CHENAB RIVER (MARALA HEADWORK)

River Chenab at Marala **Avg** Inflows in Cusecs at 0600  
27-January-2023



## KABUL RIVER (NOWSHERA)

River Kabul at Nowshera **Avg** Inflows in Cusecs at 0600  
27-January-2023







**WAPDA**

Pakistan Water & Power Development Authority

Our Vision- Dam a Decade



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## River Inflow in Pakistan

↓ Hits: 117175

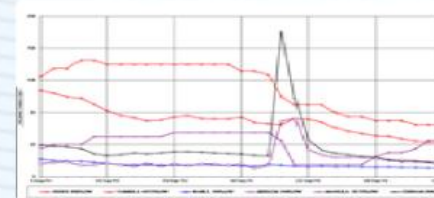


### River Flow & Levels

Flow of the river is fluctuated according to the seasons, being lower in winter, and over flowing its banks in monsoon season, from July to September 1.

[Click Here for River Flows](#)

[Click Here for Flow Below Kotri](#)



### Graphical Presentation

The principal tributaries of the Indus are the Sutlej, Beas, Chenab, Ravi, and Jhelum rivers which then merges with the Indus to form a mighty river.

[Click Here](#)

## Plans Proposed by Ministry of Water Resources, Pakistan

- Flood Forecasting and Early Warning Systems
- Radar Networks of Pakistan Metrological Departments (Existing+ Proposed)
- Meteor Bus Telecommunication System of WAPDA
- High Frequency Radio Network at Western Rivers
- Wireless Network for Eastern Rivers
- PCIW's Cross Border Hydraulic Data Sharing Plan

# SUMMARIZING THE COURSE



Database Management,  
Visualization and  
Analysis



Data Extraction, through  
Programming & Big  
Data



Optimization and  
Application



# Journals

## Journal of Hydroinformatics



ISSN

1464-7141,1465-1734



COUNTRY

United Kingdom



Scopus : Yes

WoS : Yes



SUBJECT AREA & SUB CATEGORY

- + Engineering
- + Earth and Planetary Sciences
- + Environmental Science

## Subject Area Position

ENGINEERING

EARTH AND PLANETARY SCIENCES

ENVIRONMENTAL SCIENCE

| Parameter           | Raw Score | HJRS Percentile |
|---------------------|-----------|-----------------|
| EFT <sup>TM</sup> * | 0.00176   | 61              |
| AIF <sup>TM</sup> * | 0.472     | 60              |
| SJR *               | 0.621     | 69              |
| H-Index             | 58        | 75              |
| CD2                 | 2.96      | 70              |
| SNIP *              | 1.0038    | 64              |

JPI

67

Category

W

Medallion  
Bronze

Subject Area Position  
724 / 2959

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YOUR QUESTIONS??

THANK YOU